Supplementary Information: Improving modelling for epidemic responses: reflections from members of the UK infectious disease modelling community on their experiences during the COVID-19 pandemic

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Supplementary Information

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Survey questions

How best would you describe your role in responding to COVID-19?

- Academic researcher
- Civil servant
- Professional services staff
- Other

How best would you describe the role you are doing now?

- Academic researcher
- Civil servant
- Professional services staff
- Other

Approximately how many years experience do you have in your role?

- 0-4 years experience
- 5-9 years experience
- 10+ years experience

If you feel comfortable doing so, please provide an overview of the work you did during the COVID-19 response.
Examples of this might include:

- facilitating meetings
- data management
- producing routine estimates for surveillance and submitting them to advisory bodies.
- developing tools and methods and support their use by other responders
- providing scenario estimates in response to policymakers requests
- writing reports
- academic paper writing
- attending meetings
- supporting researchers
- speaking to the media, ... etc.

On a scale from 0 to 10, how would you rate your professional experience in responding to COVID-19 in the UK?
0 - extremely negative
5 - neutral
10 - extremely positive

What professional costs, if any, did you experience from being involved in the response?

Think about both short- and long-term costs

What professional benefits, if any, did you experience from being involved in the response? Think about both short- and long-term benefits, for example:

1. number of first author papers
2. promotions/career progression
3. successful grants

On a scale from 0 to 10, how would you rate your personal experience in responding to COVID-19 in the UK?
0 - extremely negative
5 - neutral
10 - extremely positive

What personal costs, if any, did you experience from being involved in the response? Think about both short- and long-term costs

What personal benefits, if any, did you experience from being involved in the response? Think about both short- and long-term benefits

Has your employer or the wider community taken action to help mitigate any of the personal or professional costs you identified?

- Yes
- No
- Somewhat

Can you identify areas where action has been taken and areas where it has not?

Do you think there were barriers to doing effective and sustainable COVID-19 outbreak response work?

- Yes
- No
- Maybe

What were some of the barriers to doing effective and sustainable outbreak response work? For example, thinking about:

1. funders
2. employer organisations
3. supervisors
4. peers
5. computing resources
6. human resources

What can be done to reduce barriers and better support those involved in outbreak response work in the future?

What were some of the things that helped assist you to do effective research during the outbreak response?
For example, thinking about:

1. funders
2. employer organisations
3. supervisors
4. peers
5. computing resources
6. human resources

Do you think sufficient action is currently being taken to improve future outbreak responses to the standard you think is acceptable?

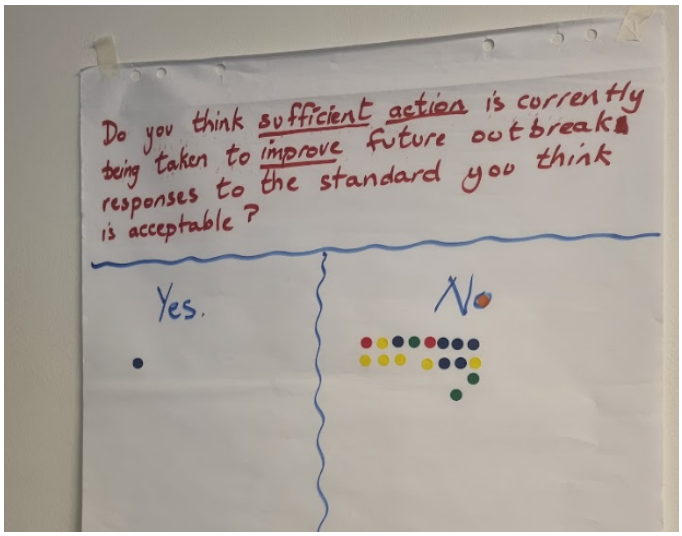
- Yes
- No
- Unsure

Is there anything else you'd like to add?

Snapshot questions

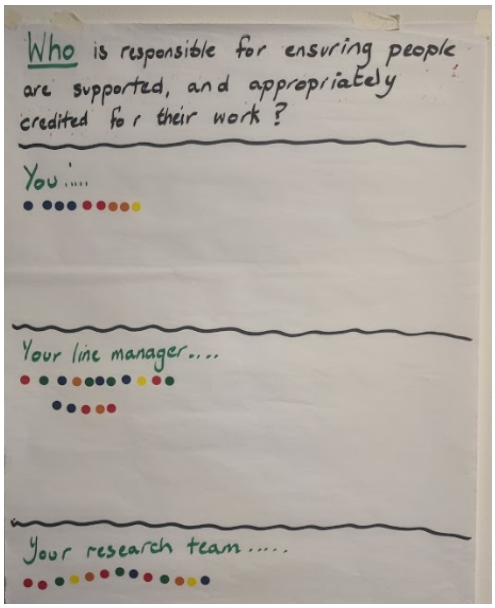
Whiteboards with the following questions were available for the first hour of the session for participants to add to with either post it notes or sticker dots. Numbers in brackets [#] represent the number of sticker dots placed by participants.

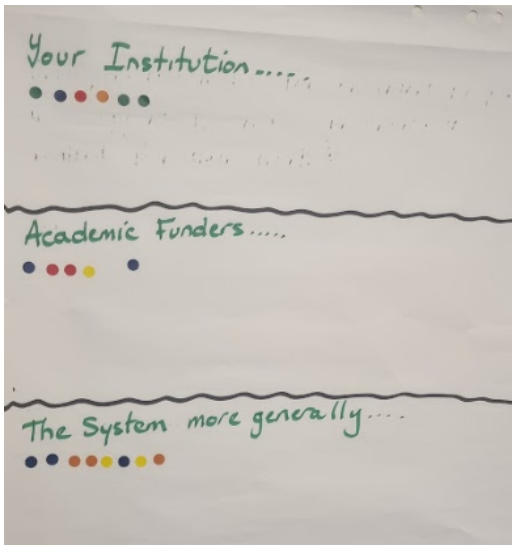
1. Do you think sufficient action is currently being taken to improve future outbreak responses to the standard you think is acceptable?



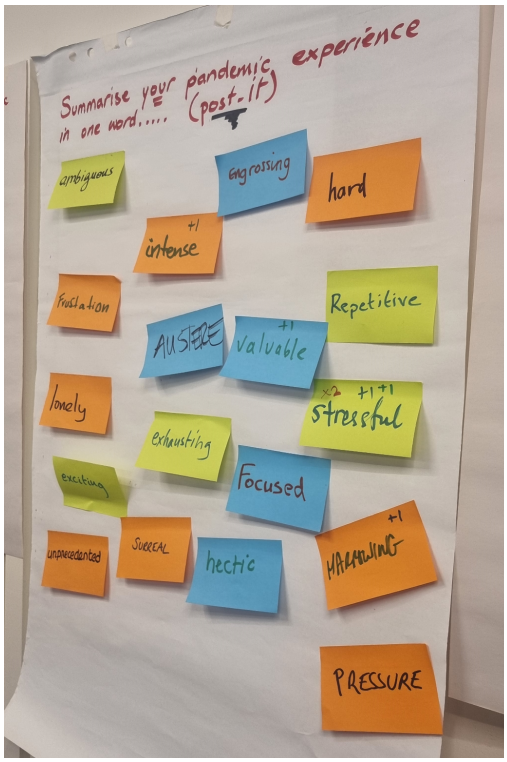
   - Yes [1]
   - No [18]

2. Who is responsible for ensuring people are supported, and appropriately credited for their work?
   - You [9]
   - Your line manager [16]
   - Your research team [13]
   - Your institution [6]
   - Academic funders [5]
   - The system more generally [8]
   - Other? [0]





3. Summarise your pandemic experience in one word.



- Ambiguous
- Engrossing
- Hard
- Intense (+1)
- Frustration
- Austere
- Valuable (+1)
- Repetitive
- Lonely
- Exhausting
- Focused
- Stressful (x2) (+1) (+1)
- Exciting
- Unprecedented
- Surreal
- Hectic
- Harrowing (+1)

Session questions

1. What was your pandemic timeline? What were the highs and lows?
2. What was your experience of pandemic work like?
   Below are some suggested directions to take this:
   - What did you expect to experience when you started working on the pandemic response, and how did this differ from what you actually experienced?
   - Did you get any benefits?
   - Were there any negatives?
   - How did your professional and personal experiences differ?
   - Do the professional gains outweigh the personal costs or vice versa?
   - Do you think privilege played a role in your experience?
3. What were some of the things that helped assist you to do effective research during the outbreak response?
   Some examples could be:
   - Management practices
   - Contract duration
   - Professional services staff
   - Funders
   - Employer organisations
   - Supervisors
   - Peers
   - Computing resources
   - HR policies
4. Do you think team science was appropriately supported over the pandemic? Some potential directions:
   - What elements of team science were most important to your work?
   - Has anything changed since prior to the pandemic in terms of support for team science?
   - What role did professional services staff, research administrators and managers, press and media teams - have on your ability to do effective response work?
5. Has your employer or the wider community taken action to help mitigate any of the personal or professional costs/challenges you identified? What more can be done?
6. Do you think there were barriers to doing effective and sustainable COVID-19 outbreak response work? If so, what were they?
7. What has been done and what more can be done to reduce any barriers to effective outbreak response work in the future

Group discussion themes

Participants formed six groups and were asked to summarise their pandemic experience using post it notes, organised into self-identified themes on a whiteboard (one per group). All text is reproduced below, source data photos available on request.

Group 1

- Life outside
    - Partner in field or not
    - Personal circumstances
    - Comradeship - shared experience
    - Importance of camaraderie in fields like military response - conflicts w/ academic reward structure
- Rewards
    - Rewards not attributed equally
    - Institutions got awards, not individuals (not all key players)
    - Don't have good mechanisms for rewarding teams (as opposed to individuals)
- Work pressure
    - Work life balance
    - Overwork
    - Lack of strong management
    - Lack of leadership
    - Line managing
    - Mismatch org - task
- Bad stuff
    - Authorship issues
    - Professional conflicts
    - Professional character assassination
- Access & privileges
    - Different layers of privilege
    - Institutional size (bigger uns = easier)
    - Country of origin
    - Well connected researcher = better access to data

Group 2

- Career direction
    - Responsibility at expense of research breadth
    - Need to get back to long term planning
    - Some career opportunities (but added stress)
    - Missing out on non-publication experiences
    - Big picture vs detail
- Work process
    - Loose in a tornado
    - Lots of competing demands

        - Difficult to stop rapid work
        - Outsourcing prioritisation
        - Hybrid work
- (Negativity about) positives
    - Shame associated with “doing well”
    - Positive ? networking + connections
    - Career advancement
    - Acknowledge previous bad work habits
- Work “feeling” / support
    - Not recognising MH impacts when you’re in it
    - Lack of understanding from non C19 colleagues
    - Missing “in person” cues in interaction
    - Need for more support - burnout
    - Lack of support / less guidance
    - Missing informal “check-ins” working remotely
    - Pride in overwork
- Personal
    - Better boundaries / priorities
    - Paused personal commitments
    - “Is it still March 2020?”

Group 3

- A
    - Team spirit
        - New vs existing
        - Lack of support
        - Individual shared
        - Competitive
        - Collaboration
    - Institutional
        - Leadership
        - Protection
        - Recognition
        - Junior members
- B
    - Mental health
        - Lack of desire
        - Reintegration
        - Uncertainty
        - Work-life balance
        - Children, partners / home, family
        - Travel
        - Guilt
        - Feeling stuck in time-space-role

        - Variable speed of time
        - Dealing with losses
    - Societal impact
        - Responsibility
        - Value of work
        - Representation of work
        - Sustainability
        - Guilt

Group 4

- COVID modeller specific
    - Everyone busy = harder to get support
    - Talking about & acknowledge the relative difficulties of each others experiences
    - Surreal level of public + media interest (good or bad)
    - Differences in levels of success & contribution resulting in inequality in various forms, i.e. lots of success based partly on luck
    - Hard to onboard and shift responsibility (sorry new people!)
    - Difficult to balance routine modelling and interesting innovative work
    - Being lumbered with “shit” jobs
    - Worked too much because too much work to do
- More general experiences
    - Worked too much because nothing else to do
    - Work + the world were one and the same - neither was an escape from the other
    - Pressure came in waves “not again…”
    - And coincided with Xmas
    - Feels like time didn’t exist i.e. no memories for a lot of it
    - Really hating the policies for young people

Group 5

- Work
    - Lack of operational data
    - What is “good enough”?
    - Not being able to say no
    - What was the point of some of the work
    - Other work pressures e.g. teaching
    - Waves of work
    - Working overnight
    - Clashes over motivation/priorities
- Emotions
    - Powerlessness
    - Sadness
    - Frustration + anger
    - Uncertainty
    - Desensitised

    - Pride
- Life
    - Skewed sense of time
    - Life events
    - Child care (x2)
    - False memories
- What to do next?

Group 6

- Structures / work
    - Shift from egalitarian structure to pre existing hierarchies
    - Collaborative work excitement
    - Team work challenges
    - Gambling / return on investment
    - All a blur
    - Survivor bias
    - Responsibility of managers to look after - who is responsible?
    - Trust
- Structures / life
    - Burn-out creeps up on you - how to avoid
    - Life events (juggling)
    - Self-care
    - Coping mechanisms
    - Personal pandemic preparedness
    - Caring for others
    - Impact of team members' personal challenges on everyone
- Emotions / work
    - Reconciling different perspectives on hierarchy & team working
    - Who gets credit
    - Space for communicating
    - Guilt
    - Comparing self with others
    - Frustration
    - Feeling useful
    - Feeling of lack of legitimacy
    - Pride
    - Self perception vs others' perception
    -
- Emotions / life
    - Trading off career vs health / everything "just 1 more"
    - Personal responsibility (feelings of)
    - Maternity leave
    - "Fresh blood" (return from leave)
    - Fear of missing out

Themes

Participants were tasked with grouping top-level themes from the group discussions into broad categories as a spider-map. This created the following:

- Work life
    - Life outside
    - Life
    - Personal
    - More general experiences
- Work - covid modeller specific
    - Work
        - Work process
        - Work pressure
    - Social impact
    - Team spirit
        - Work “feeling support”
        - Mental health
        - Emotions
- Rewards
    - Structures
    - Institutional
    - Access & priveleges
- Career direction
    - Bad stuff
    - (Negativity about) positives

Recommendations

We reproduce text from the poster boards. We include the implementor suggested by contributors at the end of each statement where this was done.

Social impact

- Educate public / policy maker of modelling knowledge for better communication
- Develop primers/training and build on existing links [*research team*]
- Ensure impactful COVID work understood at funder/institution level (esp. If less obvious/visible)
- Support for dissemination of work/case studies [*team/institution/funder*]
- Funder buy-in: creating opportunities to support consolidation of developed methods and tools
- See credit [and fund] all outputs - incl. Software and communication, media, public engagement etc. [*funders/managers/institution*]
- Formalise connections between teams/disciplines/functions - make sure we retain “what worked”
- Provide support (e.g. MH) to reduce burnout + increase retention of institutional memory
- Do[...?]ting connections [*line managers*]

Mental health

- Culture shift
    - Reframe excellence
- Raising awareness (literacy) - also of team role in crisis response
- Preparedness; peace time team building / leadership training [*wellbeing manager, centre manager*]
- Institutional safeguarding
- “We think we do this well, but we don’t” needs embedding over a sustained time
- Departmental leadership buy in; lead by example (processes - not emailing at night etc) [*funders, institution, people*]
- Resilience training (peers support) [*institution*]
- Psychological first aid *[institution*]
- Individual + management training
- Adequate resourcing of central services + research teams to mitigate against burnout = funding. [*funders, institution*]

Career direction

- Now
    - Possibly greater acceptance of portfolio PHDs
    - Leeway in examiners for PhDs during covid/emergencies
    - More recognition of non-traditional outputs
- Future
    - Improve ease of movement in/out of public health agencies (eg UKHSA), eg dual positions
    - Shifting routine data analysis to public health bodies ASAP

        - Stop/end routine activities as soon as they're not useful
        - Find a way to credit confidential work
- How/who
    - Institutions & senior academics to write letters of support for PHD students & staff with 'non-traditional' outputs [*managers, institution*]
    - Joint appointments at public health agencies [*funders*]
    - Credit all outputs: academic, tool development, communications, public engagements, confidential reporting
    - Clear expectations for promotion [*managers, funders, institutions*]
    - Distinguish "academic" vs "emergency" response
    - Disaster roster + exercise

Work process

- Capacity for cycling between response & research
- Reduce structural reliance on 1-2 people [in] a team performing specific task
- [O]n call system?
- [A]nnual leave
- Prioritise capacity
- Project manager incorporated in team [*department, funder*]
- More transparency from government committees so groups without people on them didn't get as left behind as it seemed (to me at least)
- Transparent preparedness plan *[UKHSA, govt]*
    - Who sits on gov committees
    - What their roles/responsibilities
    - White paper [*dedicated working group incl funders*]
- Clarify roles for pandemic response: software engineering, policy-related roles
- Broader reward system, e.g. [*funder*]
    - Code/software
    - reports/briefing notes
    - "Middle" authors
    - Data collection
- Automation / routine vs. one-off - value/impact?
- Regular re-assessment of cost-effectiveness of tasks
- Ask yourself if you <u>really</u> need to do this so often/ so quickly
- Weekly discussions: priorities [*team leads*]

Life/personal

- Mechanism for feeding back & instituting good working practices
- "GWP" [good working practices] reps & a committee that reports to univ./institution executive board
    - Identify people to take on the role of rep
    - Regular surveys to gauge feelings & elicit suggestions
- Time off in lieu (mandatory)
- Paid overtime (capped)

- Establish working group to make recommendations at a national level [*institution/funders*]
- Realistic funding <> deliverables
- Hobby
- Normalise taking annual leave
- Respect (enforce) working hours [*individuals & institutions*]
- Role models
- Don't try very hard (Relax)
    - Active role
- Guidelines for line managers = cultural change
- Delay emails on weekends and 7pm-7am (allow control)
    - Auto-send emails to remind people to take break from work
- PDR [performance and development review] for work/life balance (part of PDR but checked by welfare manager)
    - Palm trees on slack
    - Normalise people taking holidays > holiday snaps
- "Covid impact statement" but for more general issues
- Have paid wellbeing officer(s)/manager to check in on personal issues & balance, give professional advice, etc…

Institutional

- Credit/rewards
    - Clear comms on processes
    - Meetings non-science (coping, emotion, credits etc)
        - > mandatory incorporation in protocols; normalising [*institution*]
- Starter pack
    - Processes
    - Support & social
    - Technical
    - Mentorship/buddies
    - Available training
    - survey input regular
    - share experience between institutions [*department groups*]
- Resilience
    - Firefighter mentality
    - Back-up / replaceable
        - learn from other professions
        - Structure ahead of time & practice [*dedicated working group*]
- Responsibility
    - Hierarchy
    - Map roles
    - Back-up/replaceable
    - Function of/process for support roles (software, admin/comms)
        - Starter pack

        - available guidelines
        - structured updating [*institution, department*]
- Clear guidance
    - Communication
    - Expectations
    - Reference resource
        - Project manager
        - career path for support roles with growth
        - comms / software / admin project management [*funder, institution*]
- Manager training
    - Niche to normal
    - Timely, frequent
    - across seniority
        - Mandatory training [*institution*]
- Lessons learned
    - Exit interview feed into updating processes
    - Left since start of pandemic
    - All staff, not only academic
        - Survey / exit interview [*manager/department*]

Additional

- Pay rise
- Now now v just now prioritisation

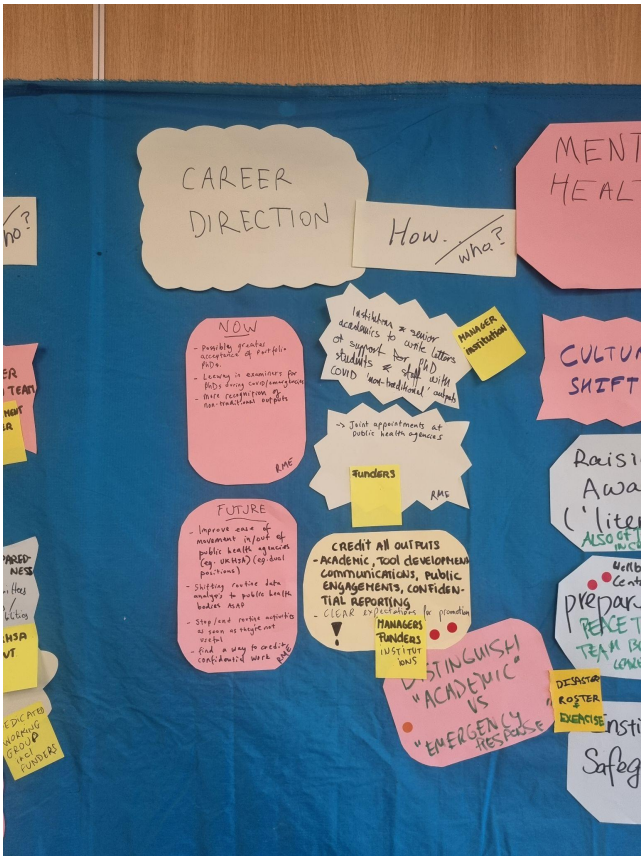


Figure 1. Example of board showing recommendations

Priority recommendations

These recommendations were highlighted by at least one participant during a group discussion. Recommendations which were highlighted by two or more participants are shown in bold.

1. Research teams should ensure that impactful COVID work is understood by funders and institutions, especially if less obvious/visible.
2. **Research teams should initiate sustainable team building and training programs during non-response periods.**
3. Employers should ensure sustainable funding for academic and professional services roles to reduce burnout risks.
4. **Employers should develop processes and guidelines for career growth support for professional services staff.**
5. **Employers need to create or expand Wellbeing Officer roles to monitor work-life balance and provide guidance.**
6. Incorporation of work-life balance components in annual performance and development reviews is essential for employers.
7. Employers should implement capped paid overtime and formal Time Off In Lieu policies and routinely analyse and act upon staff survey and exit interview data.
8. Funding bodies should adjust eligibility criteria to adequately compensate non-academic staff for activities.
9. **Funding bodies should refine impact measures to acknowledge all outputs, including academic, tool development, communications, public engagement, and confidential reporting.**